BUSINESS INTELLIGENCE (SECP3213)

ALTERYX LAB EXERCISE 3: REGRESSION & CLASSIFICATION

CHAPTER 4 (PART 3)

SEM 2, 2024/2025

STUDENT GUIDELINE

Objective:
Use Alteryx Designer to apply data investigation and time series tools to understand data patterns and forecast future trends **using predictive tool palette.**

Datasets Used
Customer_Regression_Classification.csv

Scenario:
You are a data analyst working for a retail business. Your team wants to develop models that can:

1. Predict how much customers are likely to spend based on demographic factors (regression).
2. Predict whether a customer is likely to respond to a marketing offer (classification).

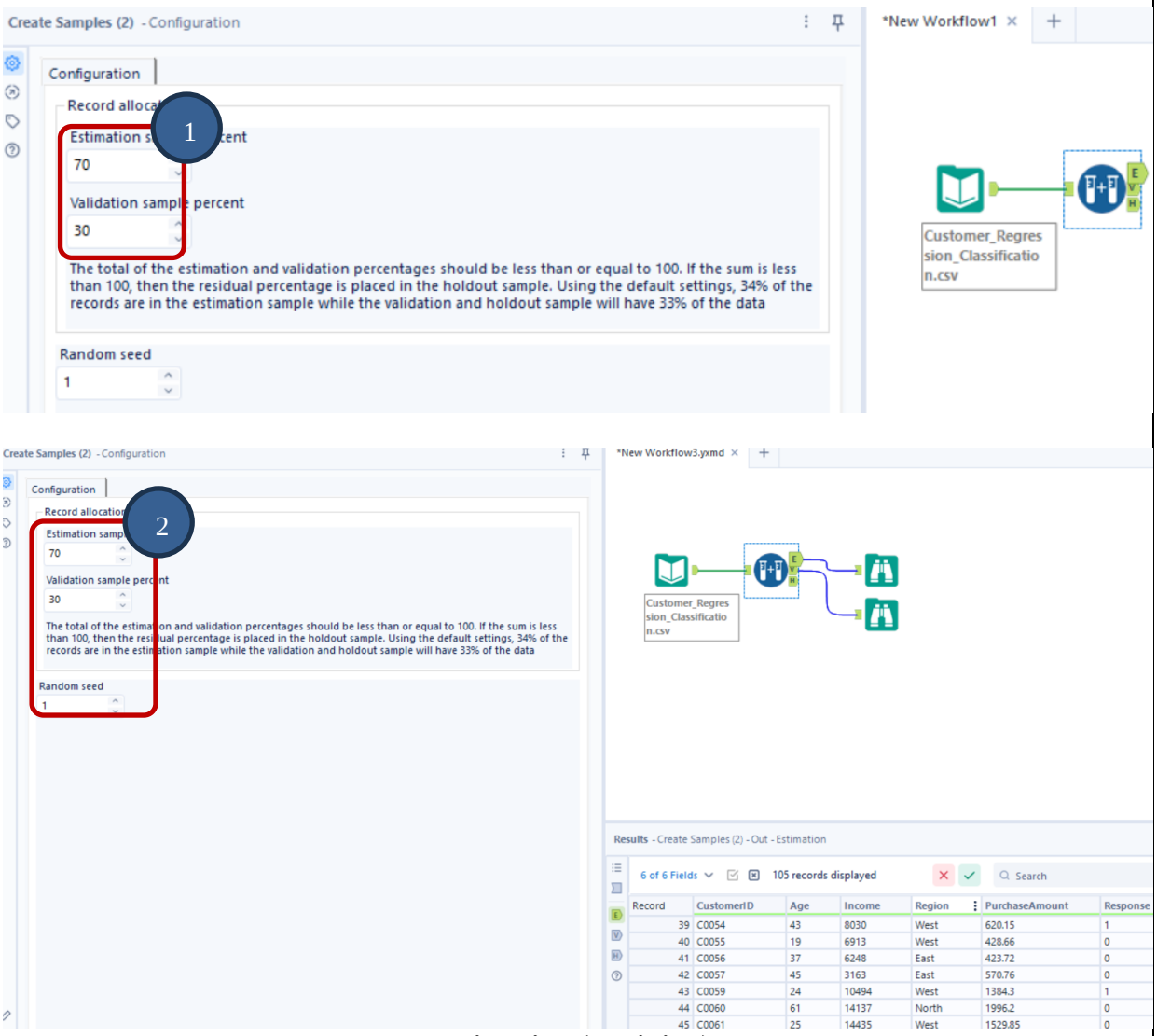
You will build, test, and compare multiple machine learning models in Alteryx to support business decision-making.

Regression Analysis

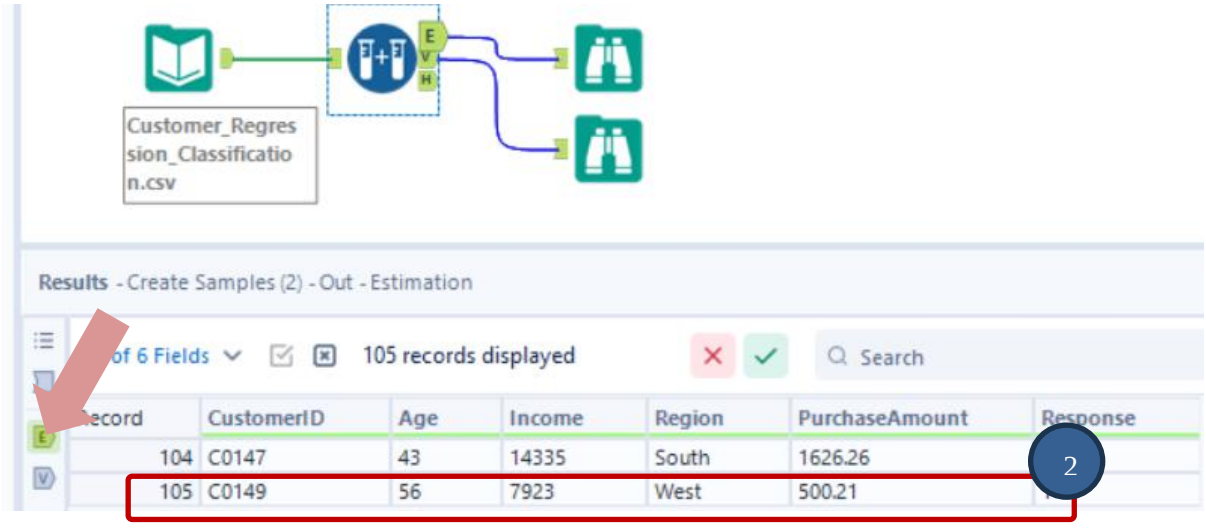
Exercise 1: Create Sample Tool

Purpose: Splits the dataset into training and validation sets to prepare for model building.

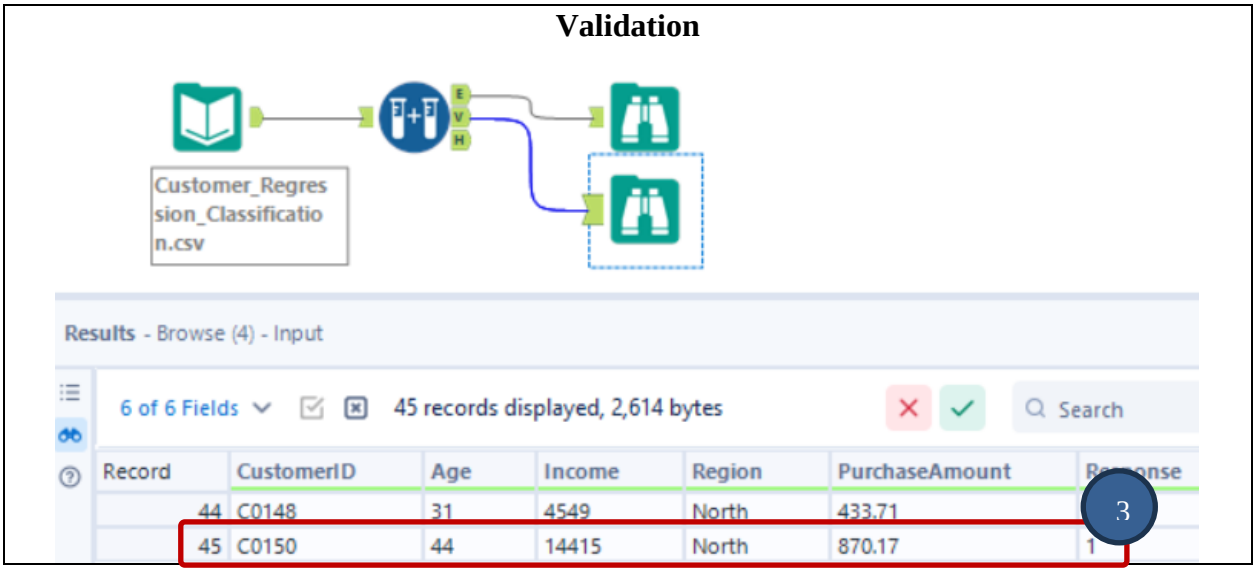
a) Use **Input Data** to load the dataset.
b) Add the **Create Samples tool** *(under preparation tool palette)*.
   - Estimation (also known as Training Set) → this is where you connect the Linear Regression or Stepwise tool.
   - Validation / Holdout → this is where you connect the Score Tool to test the model.
c) Set Sampling method to Simple Random 70/30 split. Anchor:
d) Add **Browse tools** to both outputs: Training and Validation.
   - Connect one Browse tool to the **Estimation** (**training**) output and another to the **Validation** (**holdout**) output to verify the split. You should see around 105 and 45 records respectively if using a 70/30 strategy on 150 records.
e) **Description:**
   - This approach manually splits the dataset into two parts using fixed row count.
   - The first tool extracts the first 70% of rows as the training set.
   - The second tool extracts the remaining 30% as the validation set.
   - Two datasets with ~105 and ~45 records. Browse tools show the distribution of each. Two datasets with ~105 and ~45 records.



Estimation (Training)



Validation

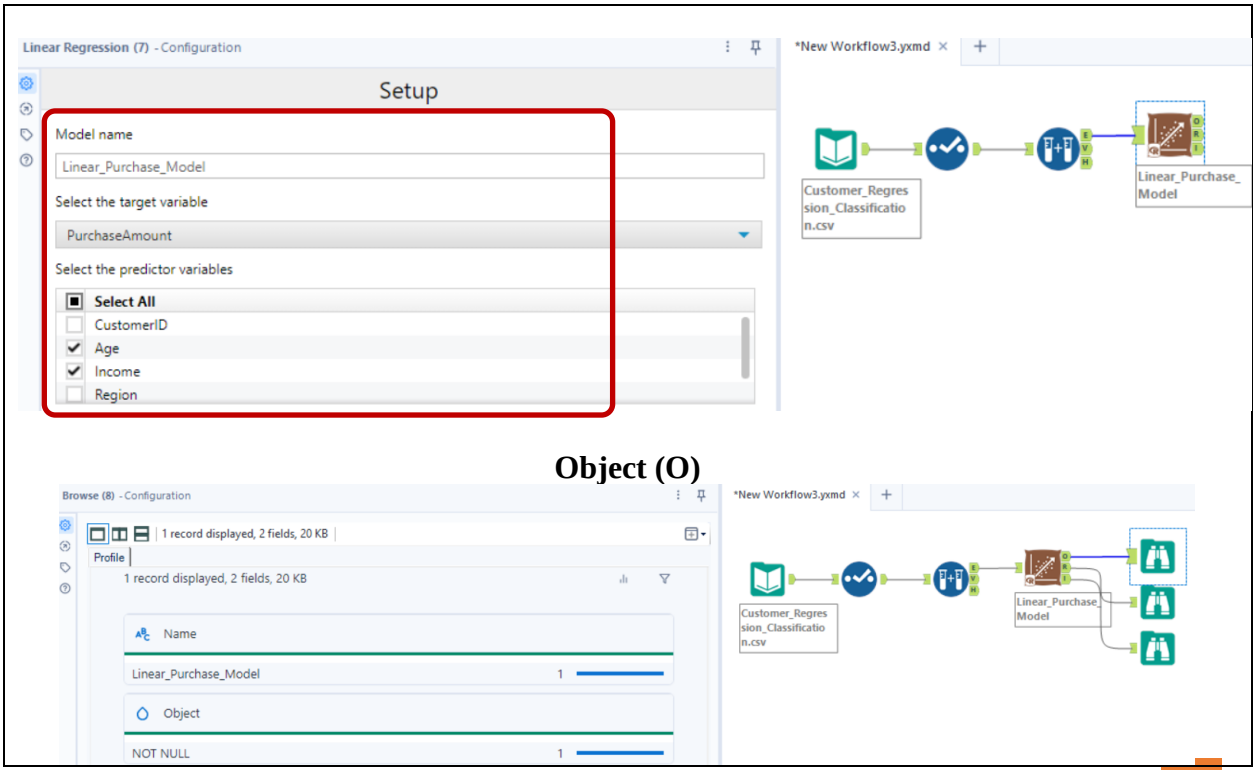


Exercise 2: Linear Regression Tool

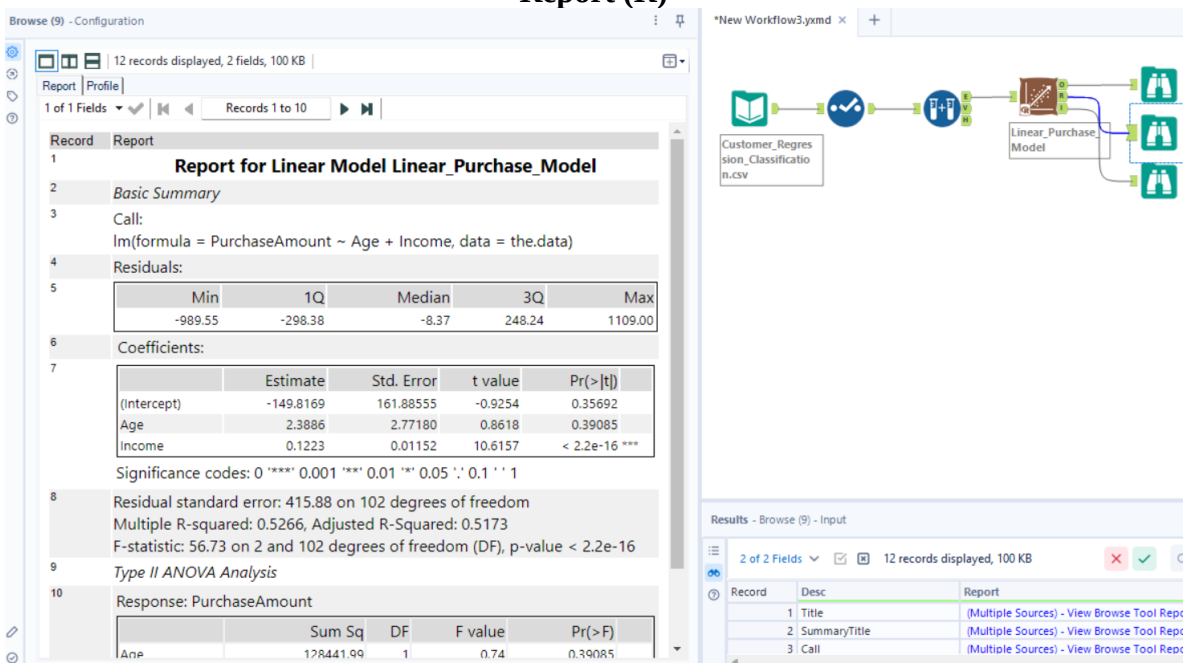
Purpose: Builds a linear model to predict PurchaseAmount using input variables.
Dataset: Output from training set.

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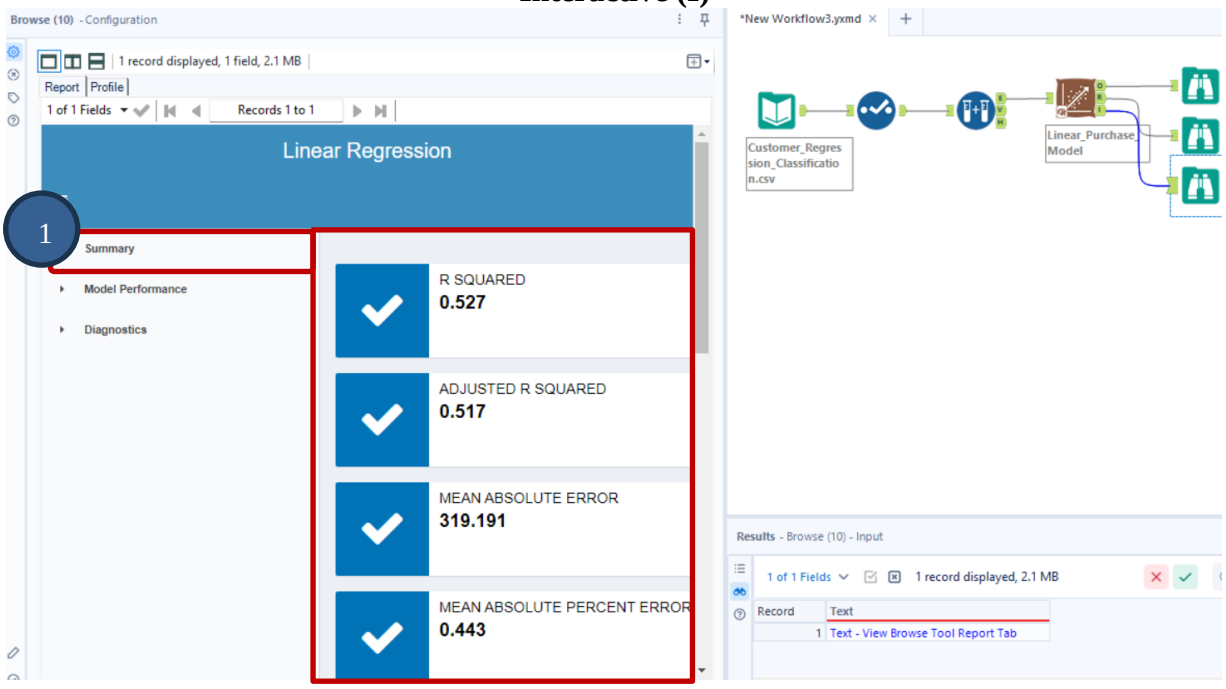
a) Connect training output to the **Linear Regression tool**.
b) Configure: Model name = Linear_Purchase_Model; Target field = PurchaseAmount; Predictors = Age, Income *(ensure that data type must be follow the correct format: PurchaseAmount → double; Age → Int; Income → double;* ***if not use select tool to change the data type****).*
c) Add Browse tool to see model summary.
d) **Description:**
   - Estimates how Age and Income influence PurchaseAmount.
   - Coefficients table and R-squared value indicating model fit.

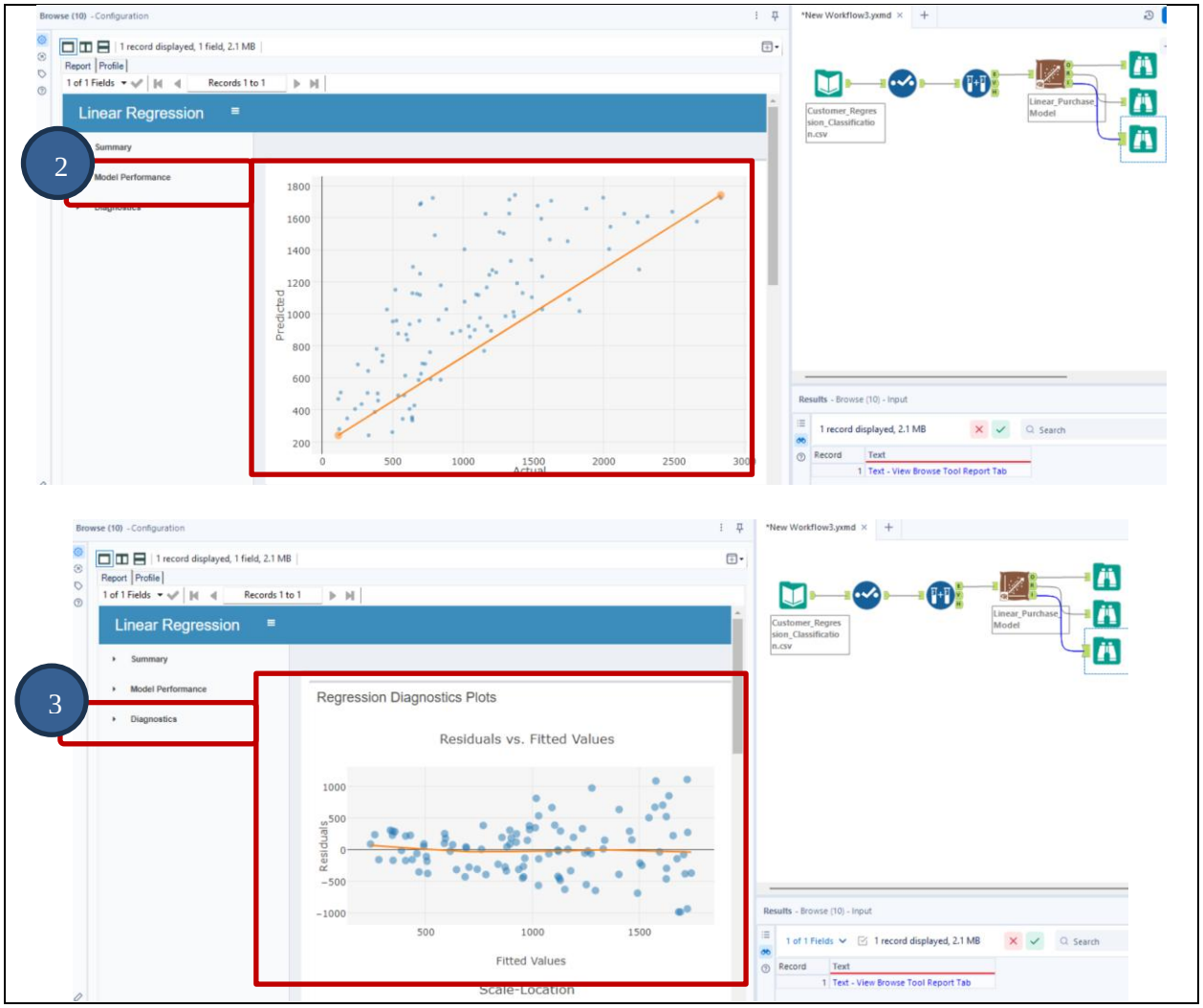


Report (R)



Interactive (I)





Exercise 3: Stepwise Tool

Purpose: Automatically selects the best subset of predictors for the regression model.

Dataset: Output from training set (Sample Tool) and Linear Regression Tool

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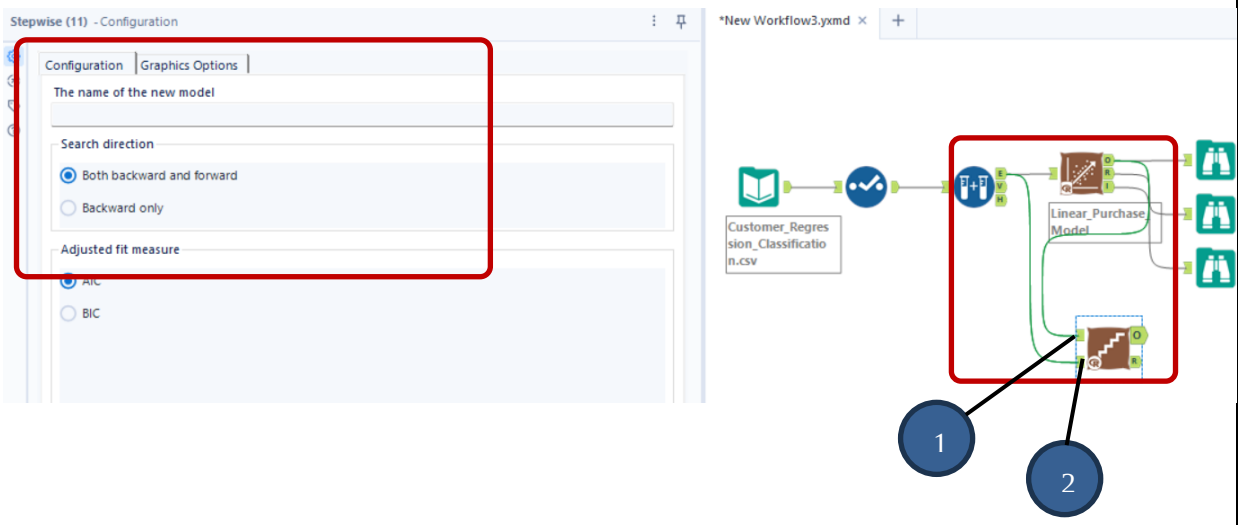
a) Connect the training dataset (from the first Sample Tool) to the **left input anchor** of the Stepwise tool. Connect from the output of Linear Regression Tool to the **right input anchor** blank (only used for updating an existing model).
   - **Left Input (Data)** – the training dataset (same data used in Linear Regression)
   - **Right Input (Optional Pre-existing Model)** – leave this unconnected if you're creating a new model from scratch

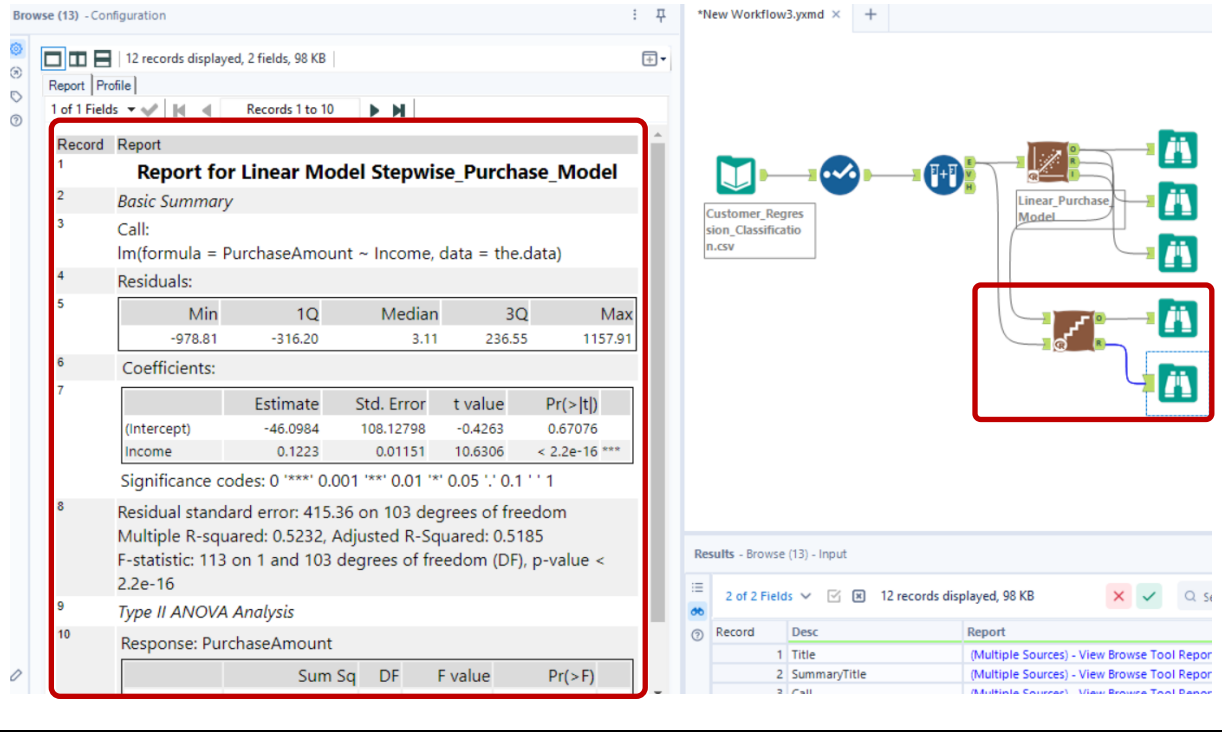
b) In configuration:
   - Model name: Stepwise_Purchase_Model
   - Search Direction: Both
   - Adjusted Fit Measure: AICc (default and recommended for small datasets)

c) Add Browse tool to see selected variables.

d) **Description**:
   - Stepwise Regression iteratively evaluates which input variables improve the model. It adds or removes predictors based on the adjusted fit measure (AICc) and direction (forward, backward, or both).
   - A regression model including only significant predictors. The output includes a list of retained variables and their coefficients.

1- **Left anchor**
2- **Right anchor**





Exercise 4: Score Tool (Regression)

Purpose: Apply the trained regression model to the validation set to make predictions.

Dataset: Output from Stepwise and validation set (Sample Tool)

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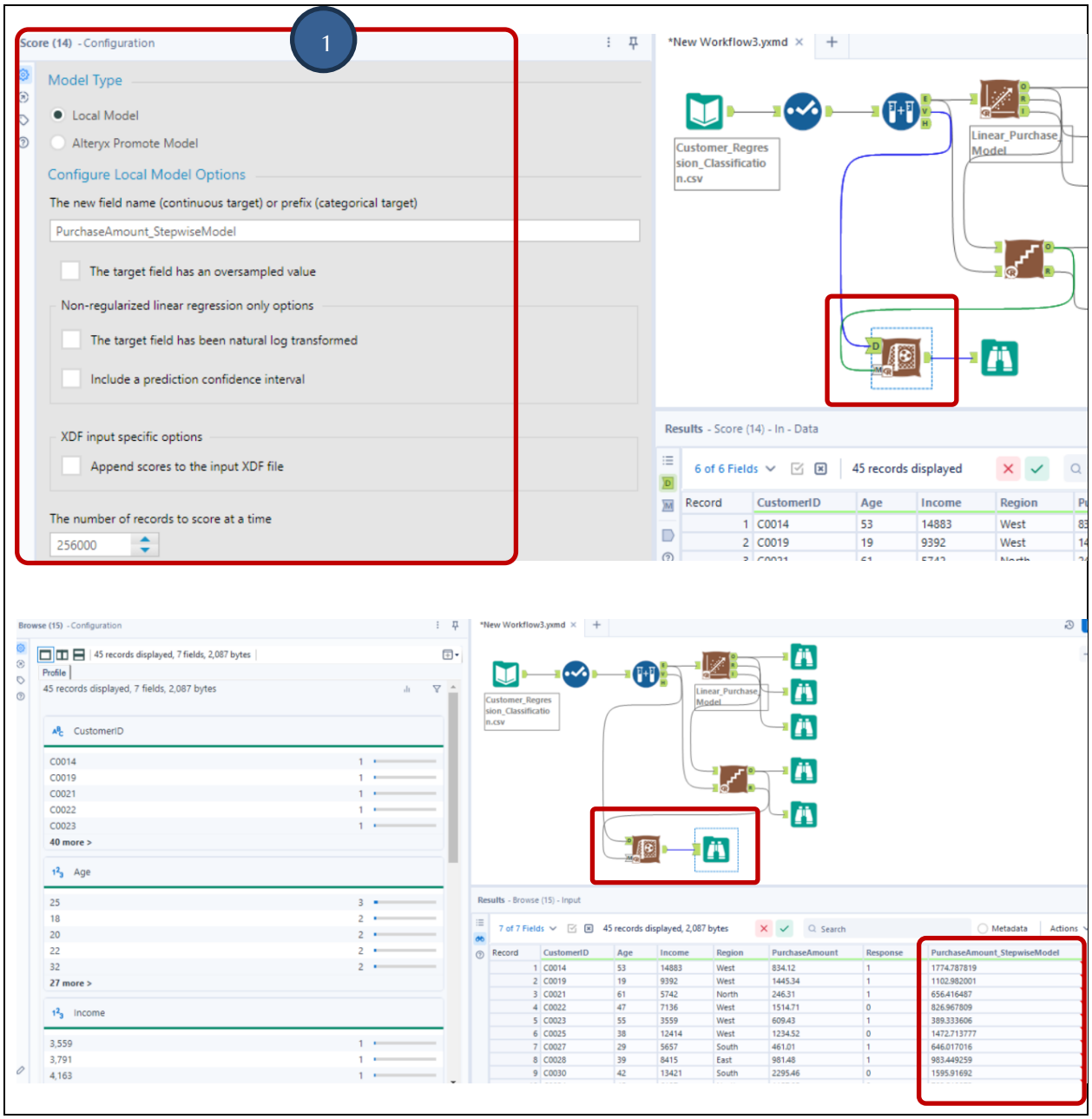
a) Connect the Stepwise output (to model anchor) and validation data (from Sample Tool to data anchor).

b) In configuration, set Local Model; New Field Name = PurchaseAmount_StepwiseModel, matching the model's name in Exercise 3.

c) Add Browse tool to see predicted PurchaseAmount.

d) **Description**:

- Used to evaluate model performance on unseen data.
- A new column with predicted values.



Classification Analysis

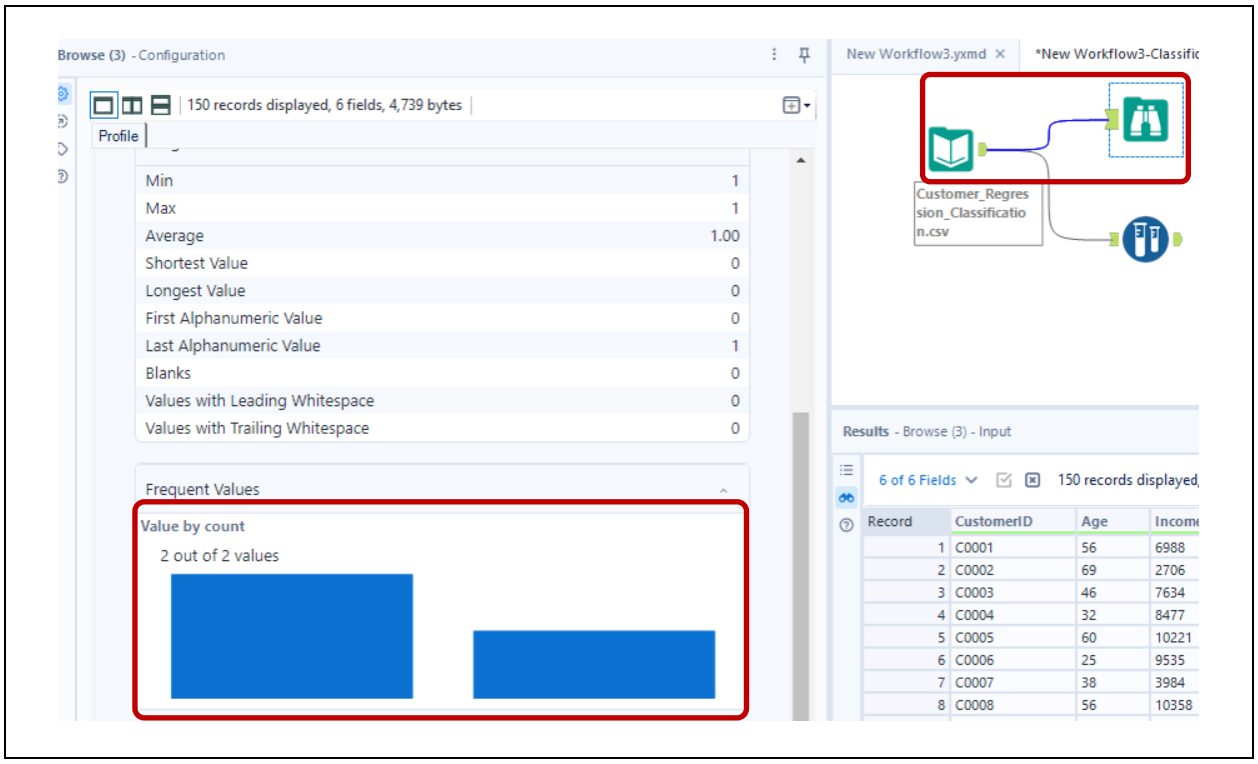
Note:
This section includes both **binary classification** (e.g., predicting whether a customer will respond = 0 or 1) and guidance that can be extended to **multiclass classification** (e.g., predicting customer segment: Bronze, Silver, Gold) if applicable.

For **binary classification**, the target field used is **Response**. To apply these steps to a **multiclass** problem, **use a categorical target field with more than two classes** (e.g., CustomerSegment) and adjust model configurations accordingly.

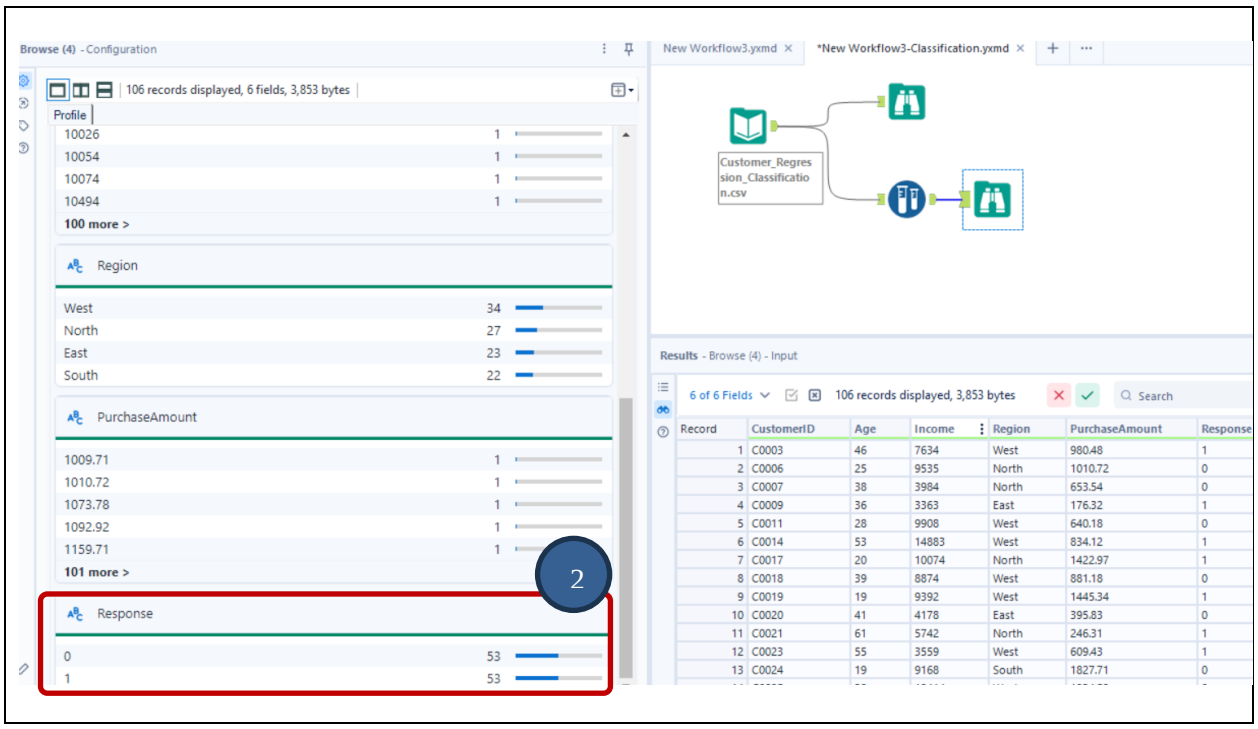
Exercise 5: Oversample Field Tool

Purpose: Balances the class distribution for binary classification problems.
Dataset: Customer_Regression_Classification.csv

a) Load the dataset and add browser to check the data if the data has unbalanced data or not. If yes proceed to add oversample field tool.



b) Add the Oversample Field tool *(preparation tool pallet).*
c) Using select tool, set age = Int; income, purchaseAmount = double.
a) Set Response as the target field and value of oversampling = 1.
b) Add Browse tool to inspect balanced dataset.
c) **Description:**
   - Useful when one class is underrepresented.
   - Answer: Dataset with approximately equal classes (0 and 1).



Exercise 8: Logistic Regression Tool

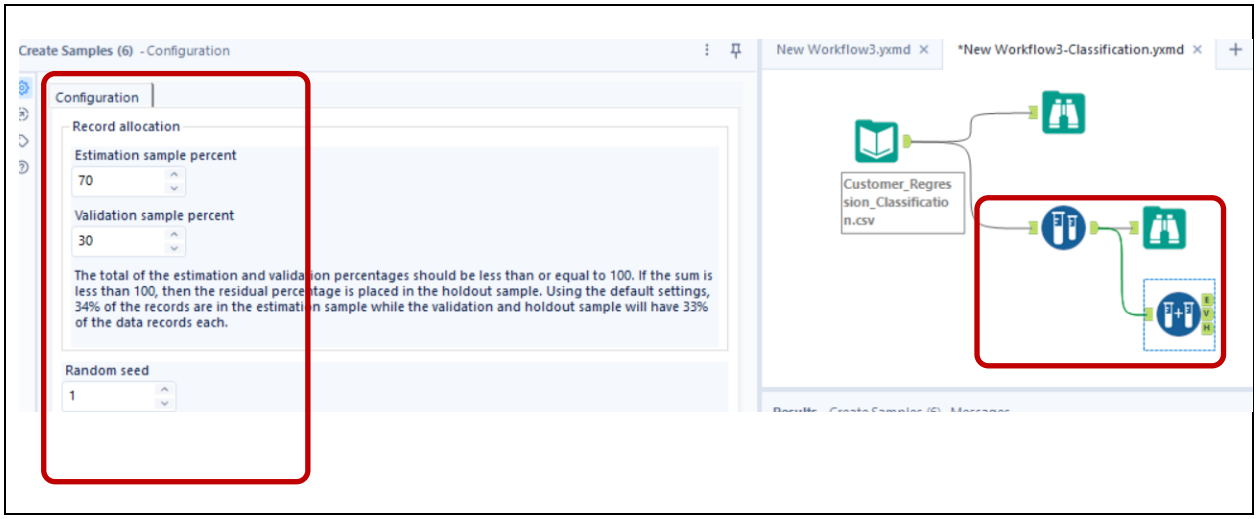
Purpose: Builds a binary classification model to predict Response.
Dataset: Balanced output from Oversample.

a) Connect oversampled data to Create Sample.
   - Set Estimation = 70; Validation = 30

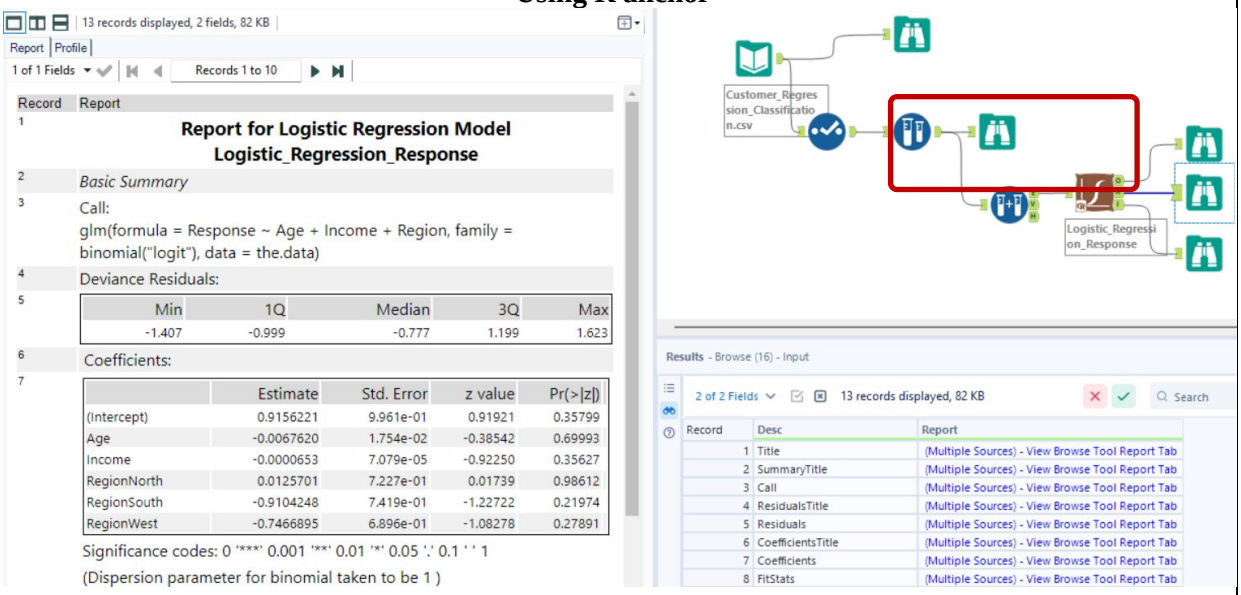
b) Connect Create Sample to Logistic Regression.
   - Connect to estimation anchor of oversampled data.
   - Set Response as target; predictors = Age, Income, Region, type model name = Logistic_Regression_Response.

c) Add Browse tool to see model coefficients and statistics.

d) **Description**:
   - Predicts likelihood of a customer responding (1).
   - Output table with odds ratios and model summary.



Using R anchor



13 records displayed, 2 fields, 82 KB

Report | Profile

1 of 1 Fields | Records 1 to 10

Record	Report
1	**Report for Logistic Regression Model** **Logistic_Regression_Response**
2	*Basic Summary*
3	Call: glm(formula = Response ~ Age + Income + Region, family = binomial("logit"), data = the.data)
4	Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.407	-0.999	-0.777	1.199	1.623

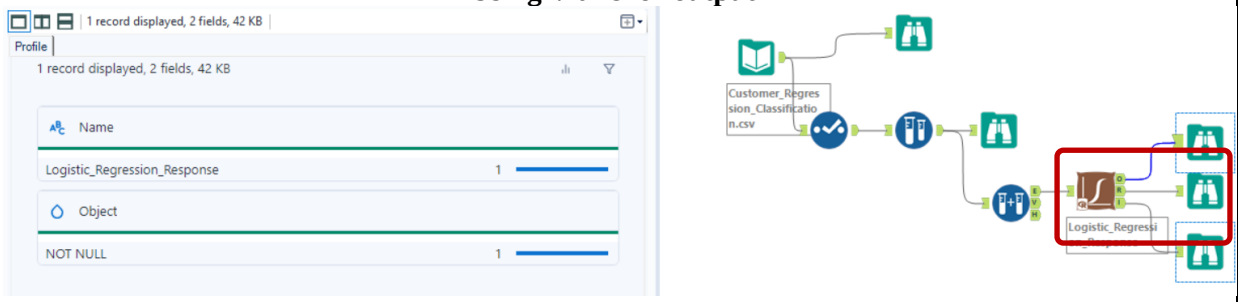
Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.9156221	9.961e-01	0.91921	0.35799
Age	-0.0067620	1.754e-02	-0.38542	0.69993
Income	-0.0000653	7.079e-05	-0.92250	0.35627
RegionNorth	0.0125701	7.227e-01	0.01739	0.98612
RegionSouth	-0.9104248	7.419e-01	-1.22722	0.21974
RegionWest	-0.7466895	6.896e-01	-1.08278	0.27891

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial taken to be 1)

Using V anchor output



1 record displayed, 2 fields, 42 KB

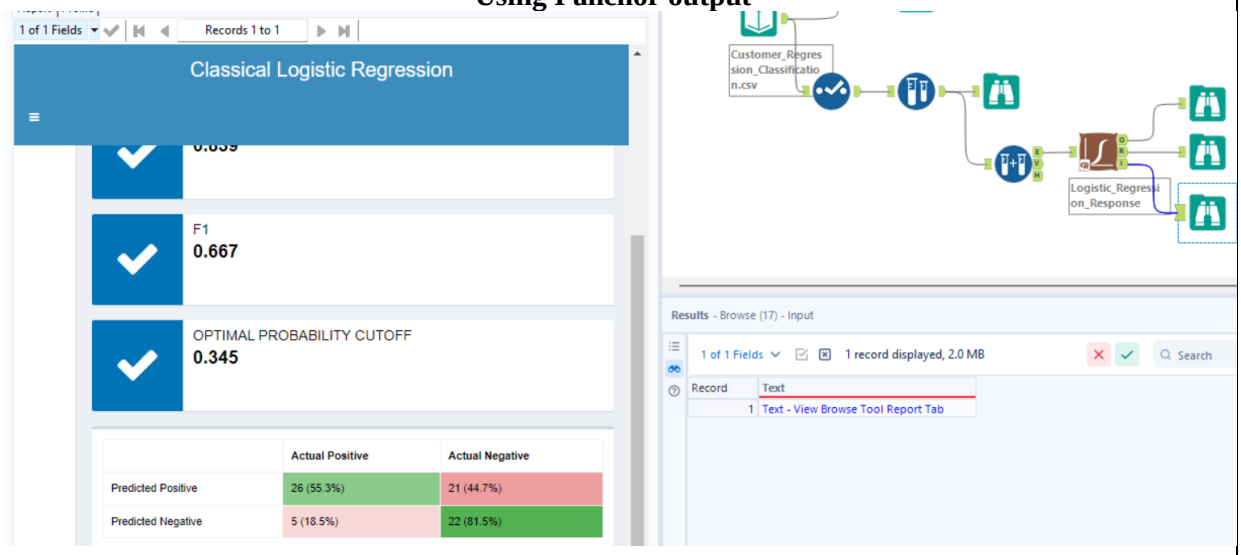
Profile

1 record displayed, 2 fields, 42 KB

Name	
Logistic_Regression_Response	1

Object	
NOT NULL	1

Using I anchor output



1 of 1 Fields | Records 1 to 1

Classical Logistic Regression

F1
0.667

OPTIMAL PROBABILITY CUTOFF
0.345

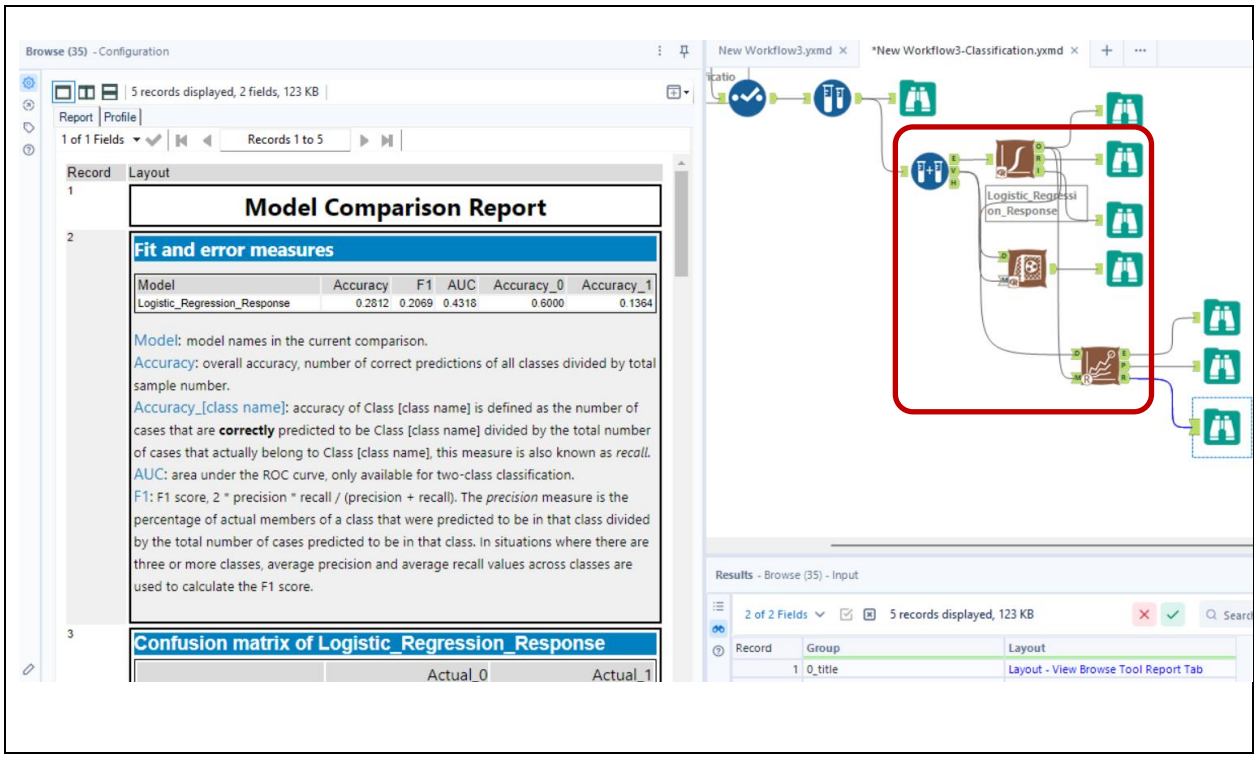
	Actual Positive	Actual Negative
Predicted Positive	26 (55.3%)	21 (44.7%)
Predicted Negative	5 (18.5%)	22 (81.5%)

Exercise 9: Lift Chart Tool
Purpose: Evaluates classification model effectiveness.
Dataset: Scored output from logistic model

a) Use the output from the Score Tool that scored the validation data using the Logistic_Response_Model.
b) Connect that scored dataset to the Lift Chart Tool.
c) In configuration:
    - Target Field: Response (the actual binary class field)
    - Score Field: The predicted probability field generated by the Score Tool (e.g., F1, Score_Response)
    - **Note**: If you encounter an error asking for a missing bottom input connection, add a Dummy Input (e.g., a Text Input tool with a single column of percentile cutoffs like 0.1 to 1.0 in increments of 0.1) to the bottom (right) anchor.
d) Add a Browse tool to the main output anchor of the Lift Chart Tool.
e) **Description:**
    - The Lift Chart helps evaluate how well the model ranks true positive cases. It compares model performance to random chance using cumulative gains.
    - Answer: Cumulative gains chart showing lift across deciles. Stronger lift in the top deciles indicates better model effectiveness in identifying responders. Cumulative gains chart showing lift for each decile segment. Higher lift in early deciles indicates better model performance. Cumulative gains chart showing lift for each decile segment. Higher lift in early deciles indicates better model performance. Cumulative gains chart.

Exercise 10: Model Comparison Tool
Purpose: Compares multiple classification models using performance metric.
Dataset: Scored output from different models

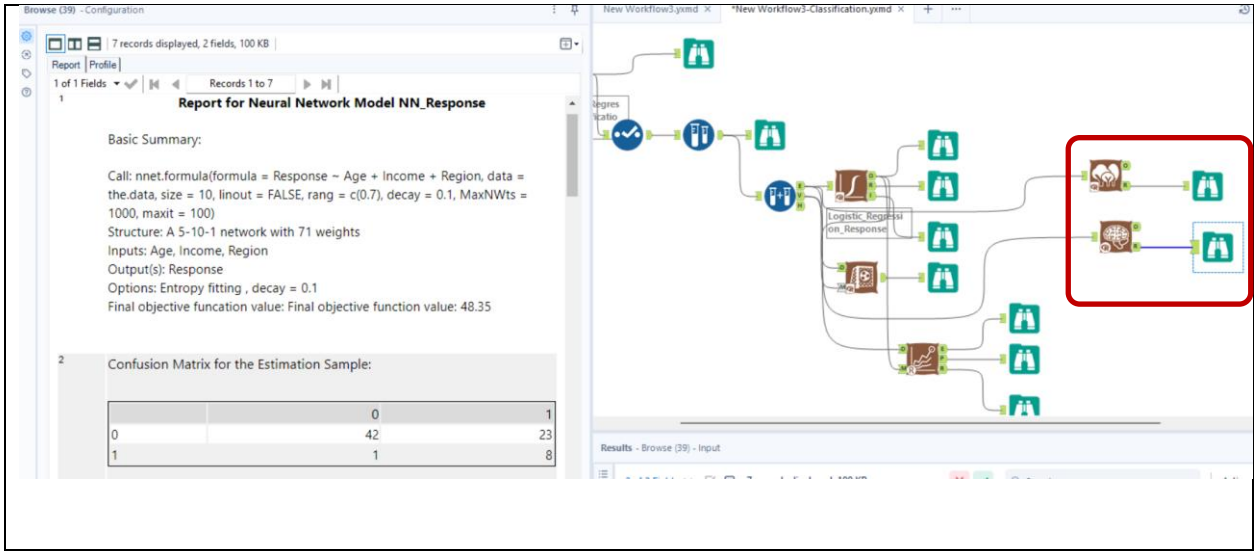
a) Action:
b) Connect logistic and create sample data validation using model comparison tools.
c) Add Browse tool to view comparison table.
d) Description:
- Ranks models based on accuracy, AUC, etc.
- Summary table with best model highlighted.



Exercise 11: Decision Tree Tool, Forest Model Tool & Neural Network
Purpose Classifies data using tree-based splitting rules.
Dataset: Balanced dataset

a) Action:
- Connect to Decision Tree Tool, Forest Model Tool & Neural Network.
- Set Response as target; predictors = Age, Income, Region.
- Add Browse tool.

b) **Description**:
- Creates interpretable rules to classify customers. Tree structure and prediction outcomes.
- Reduces overfitting by aggregating multiple trees. Confusion matrix and performance statistics.
- Captures complex, nonlinear patterns in the data. Output with predicted class labels and scores.


Browse (39) - Configuration
7 records displayed, 2 fields, 100 KB
Report
Profile
1 of 1 Fields
Records 1 to 7
Report for Neural Network Model NN_Response
Basic Summary:
Call: nnet.formula(formula = Response ~ Age + Income + Region, data = the.data, size = 10, linout = FALSE, rang = c(0.7), decay = 0.1, MaxNWts = 1000, maxit = 100)
Structure: A 5-10-1 network with 71 weights
Inputs: Age, Income, Region
Output(s): Response
Options: Entropy fitting , decay = 0.1
Final objective funcation value: Final objective function value: 48.35
Confusion Matrix for the Estimation Sample:
0 1
0 42 23
1 1 8
New Workflow3.yxmd
*New Workflow3-Classification.yxmd
Logistic_Regressi on_Response
Results - Browse (39) - Input